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ORIGINAL PAPER

Outlier SNPs enable food traceability of the southern rock lobster, *Jasus edwardsii*

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Abstract Recent advances in next-generation sequencing have enhanced the resolution of population genetic studies of non-model organisms through increased marker generation and sample throughput. Using double digest restriction site-associated DNA sequencing (ddRADseq), we investigated the population structure of the commercially important southern rock lobster, *Jasus edwardsii,* in Australia and New Zealand with the aim of identifying a panel of SNP markers that could be used to trace country of origin. Four ddRADseq libraries comprising a total of 88 individuals were sequenced on the Illumina MiSeq platform, and demultiplexed reads were used to create a reference catalog of loci. Individual reads were then mapped to the reference catalog, and variant calling was performed. We have characterized two single-nucleotide polymorphism (SNP) panels comprised in total of 656 SNPs. The first panel contained 535 neutral SNPs and the second, 121 outlier SNPs that were characteristic of being putatively under selection. Both neutral and outlier SNP panels showed significant differentiation between the two countries, with the outlier loci demonstrating much larger F_{ST} values (F_{ST} outlier SNP panel = 0.134, $P < 0.0001$; F_{ST} neutral SNP panel = 0.022, $P < 0.0001$). Assignment tests performed with the outlier SNP panel allocated 100 % of the individuals to country of origin, demonstrating the usefulness of these markers for food traceability of *J. edwardsii*.

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Introduction

Marine benthic invertebrates typically exhibit a pelagic larval phase that serves as a mechanism of dispersal and maintains connectivity between subpopulations (Eckman 1996). It is widely accepted that larvae with a short pelagic larval duration (PLD) are more prone to settling close to their parents, generating genetic structure at broad spatial scales (Palumbi 1994). In contrast, long PLDs can potentially lead to the absence of, or low population structure due to dispersal of larvae over large geographical areas (Shanks et al. 2003). However, most larval transport is largely determined by hydrodynamic features, which can cause strong genetic differentiation, even in species with a relatively long PLD (Palumbi 1994). An increasing number of studies using genetic markers have concluded that larval duration cannot be directly used as a predictor of genetic structure (Shanks 2009; Wei et al. 2013; Teske et al. 2015).

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The southern rock lobster, *Jasus edwardsii*, is distributed from southern Australia and the Tasman Sea to all coasts of New Zealand. *J. edwardsii* is a commercially important species in both countries, and fisheries management is carried out independently in Australia and New Zealand. This resource represents a substantial income for economies of both countries, providing annual revenue of approximately US$292 million to Australia (ABARE-BRS 2010) and US$204 million to New Zealand (Statistics New Zealand 2016). The main export market for both countries is Asia, where lobsters are mostly exported live (ABARE-BRS 2010; Jeffs et al. 2013).

Despite the protracted pelagic larval duration of up to 24 months (Booth and Phillips 1994), modeling simulations of larval trajectories have estimated that only 8 % of larvae released from Australia have the potential to reach New Zealand (Bruce et al. 2007). An earlier genetic study that characterized the structure of six rock lobster populations from around New Zealand and two populations in Australia found evidence for restricted gene flow across the species range (Thomas and Bell 2013). The hypothesis of a panmictic *J. edwardsii* population throughout its geographical range was rejected based on significant F_{ST} ($F_{ST} = 0.011$) from nine microsatellite markers (Thomas and Bell 2013). However, the authors suggested the possibility of larval migration from Tasmania (Australia) to central New Zealand. In support of this hypothesis, a subsequent study conducted with eight microsatellite markers using lobsters collected from Tasmania and the southern zone of the South Island of New Zealand also revealed population structure between countries (Morgan et al. 2013). The presence of genetic structure between Australia and New Zealand populations detected using microsatellites (Morgan et al. 2013; Thomas and Bell 2013) highlights the potential for using genetic markers to assign location of origin to lobsters, which may be useful for fisheries management purposes.

The use of genome-wide SNP markers, in contrast to microsatellite markers, has the potential to improve resolution in the estimation of population structure, migration rates, dispersal and population assignment (Morin et al. 2004; Benestan et al. 2015), as well as the ability to explore genomic regions under selection. Recent studies have identified high levels of population structure when analyzing small numbers of outlier markers in marine fish (Corander et al. 2013; Milano et al. 2014; Candy et al. 2015). For example, 299 neutral SNPs identified large-scale population subdivision of the widespread European hake, *Merluccius merluccius*, between the Atlantic and Mediterranean Seas, but significantly finer-scale resolution was found when analyzing just 7 and 19 outlier SNPs within the Atlantic and Mediterranean basins, respectively (Milano et al. 2014). Similarly, fine-scale population structure of eulachon, *Thaleichthys pacificus*, in North America

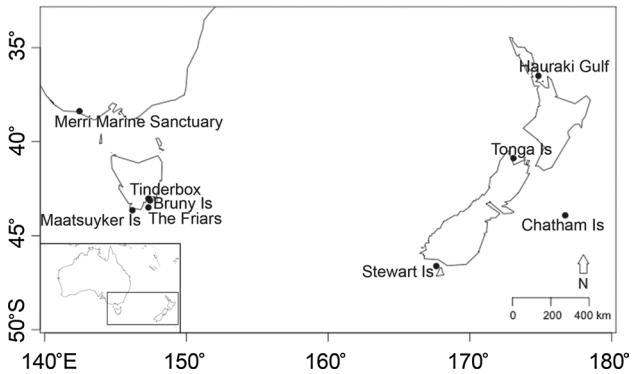

Fig. 1 Sampling sites in Australia and New Zealand

was distinguished through 193 outlier SNPs, in comparison with lower genetic differentiation detected when analyzing 3911 neutral SNPs (Candy et al. 2015). Both studies attributed the high level of genetic variation in outlier SNPs to local adaptation.

The high levels of population differentiation detected with markers under selection make them appropriate for traceability of commercial fisheries species (Araneda et al. 2016). Traceability of fish products is being increasingly used for consumer protection and for regulatory enforcement, especially in unreported and unregulated fishing (Ogden 2008). Although the *J. edwardsii* fisheries in Australia and New Zealand are managed sustainably, both countries export lobster to the Asian market. In the past, China has restricted Australian imports due to public health concerns and economic reasons. Therefore, efficient assignment of commercialized *J. edwardsii* to country of origin could prevent any conflict between Australia and New Zealand if any further bans are imposed in the future.

The aim of this study was to identify a panel of SNP markers that would enable high population assignment success and therefore could be used to trace country of origin for *J. edwardsii* to either New Zealand or Australia. We used a double digest restriction site-associated DNA (ddRADseq) approach (Peterson et al. 2012) to explore genetic structure of *J. edwardsii* using both neutral markers and markers putatively under selection. The high level of genetic differentiation exhibited by the outlier SNP panel allowed us to successfully assign individuals to population of origin.

Materials and methods

Sample collection

A total of 40 individuals from five sites (corresponding to three regions) in Australia and 48 individuals from

Table 1 Sampling sites for the *J. edwardsii* collected in Australia and New Zealand

Country	Region	Sampling site	Sampling year	Stage	Latitude	Longitude	n	n_f
Australia	Victoria	Merri Marine Sanctuary (MMS)	2013	Adult	38°23′S	142°28′E	8	7
	West Tasmania	Maatsuyker Island (MAA)	2014	Adult	43°39′S	146°12′E	11	9
	East Tasmania	The Friars (FSX)	2014	Adult	43°30′S	147°20′E	8	4
	East Tasmania	Bruny Island (BRU)	2013	Adult	43°08′S	147°27′E	4	3
	East Tasmania	Tinderbox (TXX)	2013	Adult	43°02′S	147°20′E	9	7
New Zealand	North NZ	Hauraki Gulf (HGU)	2011	Adult	36°30′S	174°50′E	13	10
	Central NZ	Tonga Island (TIS)	2013	Adult	40°53′S	173°04′E	11	11
	East NZ	Chatham Islands (CHI)	2013	Adult	43°55′S	176°43′E	12	12
	South NZ	Stewart Island (SIS)	2013	Adult	46°38′S	167°37′E	12	12

Number of individuals sequenced (n) and final number of individuals (n_f) after filtering for missing data and removal of replicates are reported for each sampling site

four sites (corresponding to four regions) in New Zealand were collected for the present study (Fig. 1) between 2011 and 2014 (Table 1). Adult lobsters on West Tasmania (AUS), East Tasmania (AUS), the Hauraki Gulf (NZ) and the Chatham Islands (NZ) were caught using commercial baited lobster pots. For the Australian samples, a pleopod clip was taken from each lobster and preserved in 90 % ethanol. For the New Zealand samples, lobster legs were removed from live specimens and frozen. Downstream analyses suggest that differences in sample preservation did not produce a batch effect, since samples from the Hauraki Gulf and Chatham Islands were assigned into the same cluster as the rest of New Zealand sampling sites (see Results). In the case of Stewart Island (NZ), Tonga Island (NZ) and Merri Marine Sanctuary (AUS), adult lobsters were collected by divers and legs (New Zealand specimens) or pleopod clips (Australian specimens) were taken from each lobster and immediately preserved in 90 % ethanol.

DNA extractions

DNA was extracted using the DNeasy Blood and Tissue kit (Qiagen) following the manufacturer's instructions. The DNA concentration of each sample was determined using a Qubit® 2.0 Fluorometer (Life Technologies). DNA integrity was determined through gel electrophoresis, and samples with predominantly high molecular weight DNA (corresponding to a band 1000 base pairs, bp, or higher) were preferentially selected for ddRADseq library preparation.

ddRADseq library preparation and sequencing

A modified version of the ddRADseq protocol developed by Peterson et al. (2012) was used to make multiplexed sequencing libraries (https://molecularbiodiversity.wordpress.com/home/protocols/). Briefly, 250 ng of genomic DNA was digested using the restriction enzyme EcoRI (GAATTC, infrequent cutter) and AciI (CCGC, frequent cutter) at 37 °C for 16 h. Subsequently, sequencing adapters containing in-line barcodes were ligated to the sheared DNA fragments, after which low molecular weight DNA fragments as well as non-ligated adapters were removed using a double size selection protocol employing Agencourt AMPure XP magnetic beads (Beckman Coulter) (Lennon et al. 2010). Index sequences (based on TruSeq LT) and flow cell attachment regions were incorporated by primer extension PCR. PCR products were cleaned using AMPure XP (Beckman Coulter), after which the DNA concentration was standardized, pooled and a gel size selection was performed to obtain a DNA fragments between 400 and 500 bp. DNA was extracted from the gel using the Wizard SV Gel and PCR Clean-Up System (Promega), and the concentration of the final ddRADseq library was determined using Qubit 2.0 fluorometer. All ddRADseq libraries were spiked with 10 % PhiX Control v3 and sequenced on the Illumina MiSeq next-generation sequencing platform using v2 2 × 250 bp kits.

The number of samples to be sequenced in a single run was determined through two pilot sequencing runs. For the first pilot run, 12 individuals from New Zealand were sequenced, and for the second pilot run 13 specimens from Australia were sequenced. We determined the number of polymorphic loci and the number of reads and sequencing depth per individual and concluded that ddRADseq libraries consisting of 45 individuals would yield sufficient coverage and depth of loci to be sequenced in a single run. Subsequently, two more ddRADseq libraries were sequenced, for a total of four ddRADseq datasets.

Technical replicates

Technical replicates were included in the second, third and fourth sequencing libraries prepared. This was particularly important given that our ddRADseq libraries were

not prepared and sequenced at the same time, which may have had introduced technical- and sequencing-derived differences between libraries (Mastretta-Yanes et al. 2015). A Principal component analysis (PCA) was used to visualize the spatial distribution of replicates and identify whether there was a batch effect due to library preparation and sequencing that could bias results. The PCA was performed using the R package Adegenet v.1.4-1 (Jombart and Ahmed 2011). A summary of the distribution of technical replicates among libraries is given in Table S1.

Preliminary analyses of raw sequencing data

Quality of the reads was initially examined using the FastQC v.0.10.1 quality control tool (Babraham Bioinformatics). Subsequently, uniquely indexed and bar-coded samples were demultiplexed using the "process_radtags" protocol from Stacks v.1.29 (Catchen et al. 2011). Based on the FastQC report, sequences were trimmed to 75 bp to assure that the Phred quality score (Q) of all reads was above 30. Trimmed reads were assessed for bacterial and viral contamination using Kraken v.3.5.0 (Wood and Salzberg 2014). This software compares sequence reads against a database to identify reads that match the taxonomic groups present in the database. Reads that do not match those of the database are output as "unclassified," or non-bacterial or viral reads, which were extracted from the raw trimmed data for further analysis. In the absence of a reference genome, removal of bacterial or viral reads is important since it is impossible to determine whether a sequence belongs to the study organism or to a contaminant, providing biased results in downstream analyses (see Merchant et al. 2014).

Reference catalog building, alignment and variant calling

Since *J. edwardsii* is a non-model species, we identified a catalog of the most frequently sequenced loci using the "rad-loci" pipeline (https://github.com/molecularbiodiversity/rad-loci) developed at La Trobe University. Sequence data from all individuals were first pooled and then clustered using VSearch v.1.1.3 to identify putative loci. Initially, only clusters of reads with a depth of at least 103 (and therefore likely to appear at least once in most of the samples) were retained. This means that the number of raw reads required to form a cluster was 103. Sequences were considered to be sufficiently related if they shared at least 94 % identity (4 bp maximum difference in a 75 bp read, allowing single-nucleotide variations and indels). Therefore, the allowed number of mismatches between two clusters was set to 5 bp. Assuming that each member of the cluster was an allele, only clusters that were composed by a minimum of two members and a maximum of 16 members were kept. The minimum number was based on the fact that we wanted to obtain bi-allelic data. A second round of clustering of the remaining reads at 94 % identity was performed, followed by another filtering of clusters that were not comprised by a minimum of two and a maximum of 16 members. After re-filtering, individual samples were mapped back to the filtered clusters, each cluster was now called a "locus" and it consisted of one representative sequence and up to 16 variations. Finally, samples were mapped back to the identified loci allowing for a maximum of 40 % missing data across samples in each locus. The total number of "reference" loci identified by this pipeline was 1054 (Table S2).

Individual reads were subsequently mapped to the reference loci catalog using the software Bowtie2 v.0.7.12 (Langmead and Salzberg 2012). Finally, variant calling of mapped loci was performed using the Genome Analysis Toolkit (GATK) v.3.3_0 (McKenna et al. 2010), yielding a total of 947 SNPs (Table S2).

SNP filtering

The putative RAD loci were filtered to ensure that known confounding variables, such as non bi-allelic loci, missing loci, allele dropout (ADO) and linkage disequilibrium (LD) (Henning et al. 2014), were minimized prior to population level analyses using VCFtools v.0.1.13 (Danecek et al. 2011). Only bi-allelic loci were retained, using the options—min-alleles 2—max-alleles 2. Additionally, we selected a single SNP per locus with the option—thin 75, given that loci were 75 bp long. Since paired-end sequencing was performed, and therefore, pairs of loci of 75 bp length could potentially be in LD, we set a pairwise LD measure threshold of $r^2 > 0.8$ to remove potentially linked loci using the option—min $- r^2$. Average coverage was 44, allowing for a minimum coverage of 5× to minimize ADO. Rare alleles were also removed by setting a minor allele frequency of 0.1 with the option—maf. Finally, both loci and individuals with more than 20 % missing data were excluded from the analysis, yielding a final total number of 656 SNPs (Table S2).

SNP characterization

Detection of neutral loci and loci putatively under selection was performed using Lositan (Beaumont and Nichols 1996; Antao et al. 2008) using 10 x 10^4 simulations, a confidence interval of 0.99 and a false discovery rate of 0.1 (Jacobsen et al. 2014). Lositan uses an F_{ST} outlier approach that identifies loci as outliers when their F_{ST} values are too high or too low compared to neutral expectations (Antao et al. 2008). Loci found to be under balancing and positive

selection will be hereafter referred as outlier loci. All downstream analyses were performed and are reported for each SNP panel separately.

Finally, to examine whether sequence reads aligned to protein coding regions, demultiplexed untrimmed reads were screened through tBLASTx v.2.2.29+ (Altschul et al. 1997). This program searches a translated nucleotide database with putative translated nucleotide queries. Raw reads (150 bp) were used rather than the 75 bp reference loci to improve the BLAST alignment length and hence specificity. Subsequently, queries with statistically significant *e* values ($E < 0.01$, Karlin and Altschul 1990) were screened against the reference loci using BLASTn v.2.2.29 +(Altschul et al. 1997) to identify any loci that were contained on those reads and therefore linked to those genes.

Analyses of genetic diversity

The level of observed (H_o) and expected heterozygosity (H_E) in each population, as well as *F* statistics for both SNP panels, were calculated using the R packages Adegenet v.1.4-1 (Jombart and Ahmed 2011) and Pegas v.0.8-2 (Paradis 2010). A two-sample Wilcoxon test was used to detect whether mean H_o differed significantly from mean H_E for both SNP panels.

Effective population size estimation

Effective population size (Ne) was estimated by the software NeEstimator v.2.01 (Do et al. 2014) using the linkage disequilibrium model based on allele frequencies of all 656 loci. This model provides the most precise estimation of Ne among other single-sample methods (Waples and Do 2010; Do et al. 2014).

Analysis of population structure

Population structure was investigated through discriminant analysis of principal components (DAPC). DAPC was performed, and results were plotted using the R package Adegenet v.1.4-1 (Jombart and Ahmed 2011). This analysis assigns individuals to clusters and selects the best number of clusters based on Bayesian information criterion (BIC) (Jombart et al. 2010). The results are comparable to those obtained by STRUCTURE (Pritchard et al. 2000), with the advantage that DAPC explores genetic structure without making assumptions about the genetic model of the study population (Jombart et al. 2010).

Assignment of individuals to country of origin

Assignment tests were performed using the program Geneclass2.0 (Piry et al. 2004) to test the effectiveness of the outlier SNP panel to assign individuals to country of origin. We simulated 10×10^3 multilocus genotypes using the algorithm described by Paetkau et al. (2004). We used the Bayesian allele frequency estimation method (Rannala and Mountain (1997) whereby each individual is removed from the baseline and assigned to the most likely population.

Results

Variant calling and SNP filtering

ddRADseq sequencing libraries were prepared from seven subpopulations of the southern red rock lobster from Australia and New Zealand and sequenced using an Illumina MiSeq. A "reference" catalog of loci was developed from a total of 88 individuals using an average of 580,000 reads per individual. After variant calling and SNP filtering, our final dataset comprised 75 individuals, including 30 Australian and 45 New Zealand samples. From this data, a total of 954 SNPs were identified, and after filtering (MAF = 0.1, LD < 0.8 between loci, 1 SNP per loci, <20 % missing data of loci and individuals), a total of 656 SNPs were obtained. Using the software Lositan to discriminate between putatively neutral variants and variants characteristic of being under selection, two panels of SNPs were identified consisting of 535 neutral and 121 outlier SNPs, respectively.

Consistency in assignment of technical replicates

Technical replicates were included in three ddRADseq libraries. The PCA showed consistency in assigning technical replicates from the same individual close to each other and within the cluster of each country. This means there was no batch effect produced by different library preparation and sequencing runs (Fig. S1).

Genetic diversity and effective population size

Mean expected and observed heterozygosity for both countries were higher for the neutral SNP panel than for the outlier panel and the New Zealand population exhibited higher heterozygosity than the Australian population (Table 2). The negative values of the inbreeding coefficient (F_{IS}) estimated for both SNP panels and both countries are indicative of heterozygote excess. The two-sample Wilcoxon test indicated that H_O was significantly higher than H_E in both SNP panels for the New Zealand population ($P < 2.2e - 16$ and $P = 0.0199$, respectively, for the neutral and outlier SNP panels). For the Australian population, significant differences between H_O and H_E were only found for the neutral SNP panel ($P < 2.2e - 16$ and $P = 0.07222$, respectively, for the neutral and outlier SNP panels).

Table 2 Descriptive statistics for Australia and New Zealand given by the neutral and outlier SNP panels

	N	N_{loci}	H_O (±SD)	H_E (±SD)	F_{IS}	F_{ST}	P
Neutral SNP panel							
New Zealand	45	535	0.633 (±0.279)	0.399 (±0.118)	−0.588		
Australia	30	535	0.518 (±0.268)	0.369 (±0.126)	−0.401	0.022	<0.0001
Outlier SNP panel							
New Zealand	45	121	0.386 (±0.319)	0.264 (±0.176)	−0.463		
Australia	30	121	0.355 (±0.271)	0.273 (±0.169)	−0.297	0.134	<0.0001

N sample size, N_{loci} number of SNPs, H_O mean observed level of heterozygosity, H_E mean expected level of heterozygosity, F_{IS} fixation index (inbreeding coefficient), F_{ST} pairwise fixation index between Australia and New Zealand

Estimation of effective population size in Australia and New Zealand gave infinite values, with confidence intervals of 1334.6 − ∞ for Australia and infinite for the New Zealand *J. edwardsii* population. Infinite values of estimated Ne and confidence intervals may be due to larger than expected sampling error (Do et al. 2014).

Population structure

While both panels detected population structure between Australia and New Zealand, the outlier loci detected greater differentiation between countries (Table 2). DAPC analysis using both SNP panels detected two clusters, each of them representing each of the two countries (Fig. 2). However, the higher genetic divergence given by the outlier SNP panel was detectable in the distance between both clusters, which was one order of magnitude higher than that of the neutral SNP panel (Fig. 2a, c, respectively). To determine the outlier SNPs causing most of the differentiation between countries, we inspected the associated allele loadings (Jombart et al. 2010), showing that 30 SNPs contribute to the discrimination between Australia and New Zealand (Fig. S2).

Assignment of individuals to country of origin

The outlier SNP panel correctly assigned 100 % of individuals to population of origin, with a quality index of 87.72 % (Table S3).

Characterization of catalog loci

An analysis of the ddRAD loci to identify coding sequence variants using tBLASTx did not produce any significant alignment with loci containing neutral or outlier SNPs.

Discussion

This study represents the first genome-wide population genetic analysis of a rock lobster species. We produced a panel of outlier and neutral loci to investigate population structure between Australia and New Zealand and the potential for assignment of individuals to population of origin. The southern red rock lobster, *J. edwardsii*, is a commercially important species in Australia and New Zealand, and given reported evidence of differing degrees of genetic connectivity (Morgan et al. 2013; Thomas and Bell 2013), further fine-scale genetic mapping of populations from both countries is required to understand the extent of connectivity between countries. As seen in other marine species, our outlier SNP panel showed greater genetic differentiation between Australia and New Zealand than the neutral panel. Therefore, we propose that the outlier SNP panel has potential as an effective method for determining population of origin in *J. edwardsii*.

Evidence for a population bottleneck

Significant differences between mean H_O and H_E were detected for the neutral SNP panel in both countries and the outlier SNP panel in New Zealand, and the negative values of the fixation index indicate heterozygote excess. Heterozygote excess can be a result of low depth of coverage, since in the presence of low depth data homozygote genotypes could be erroneously called heterozygous (Nielsen et al. 2012b). However, we do not consider this to be the case in the present study since our average depth of coverage is 43 and it has been suggested that a minimum of 5× coverage gives reliable estimates of allele frequencies (Ellegren 2014).

Heterozygote excess may also be the result of a small effective population size caused by a past population bottleneck many generations ago (Luikart and Cornuet 1998) or as a result of a relatively small number of individuals contributing to each generation (Hedgecock and Pudovkin 2011). However, heterozygote excess caused by a population bottleneck can be a transient state (Luikart and Cornuet 1998). Thomas and Bell (2013) estimated an effective population size of 2717 in Australia and 1189 in New Zealand. These estimates are low, and therefore, it is possible that the

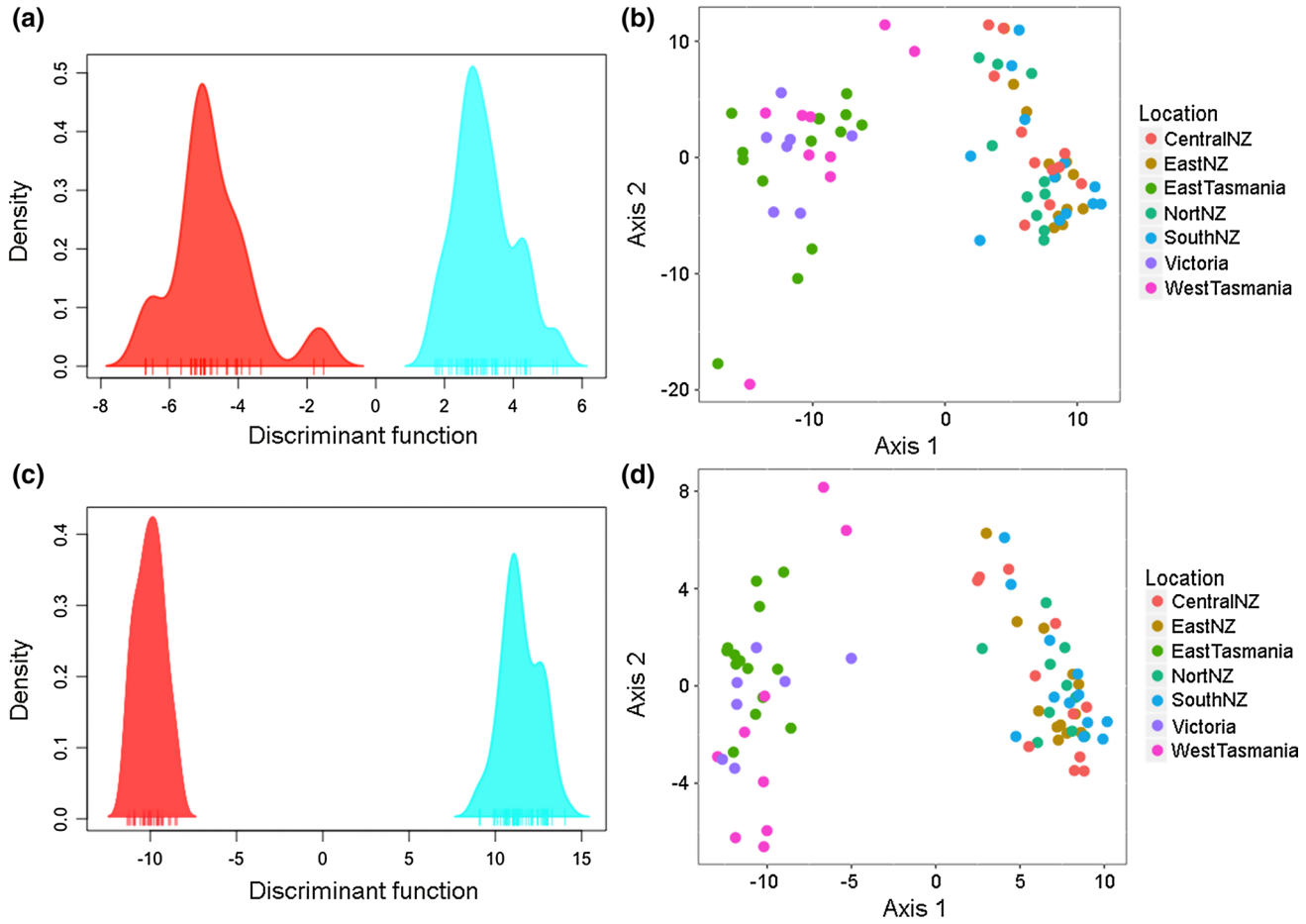

Fig. 2 First principal component resulting from the discriminant analysis of principal components (DAPC) using **a** the neutral SNP panel and **c** the outlier SNP panel. The *red* group represents 30 Australian individuals, while the *blue* group represents 45 New Zealand individuals. Principal component 1 and 2 resulting from a principal component analysis (PCA) using **b** the neutral SNP panel and **d** the outlier SNP panel. Sampling regions are represented by *colors*

populations have undergone a bottleneck. The infinite estimates of Ne in the present study do not provide any evidence of genetic drift due to small Ne (Do et al. 2014). However, as discussed above, our infinite values of estimated Ne and confidence intervals may be due to larger than expected sampling error. When Ne is large or if there are limited data available, the estimate of Ne will be negative and the biological interpretation is that Ne $= \infty$ (see Waples and Do 2010).

Performance of the neutral SNP panel for detecting population structure

Studies on population structure of marine organisms based on neutral markers typically demonstrate significant but weak genetic differentiation (Nielsen et al. 2009; Milano et al. 2014). Neutral markers are affected by demography and the evolutionary history of populations; therefore, they evolve as a result of genetic drift and migration (Luikart et al. 2003). For this reason, even low levels of migrant exchange can maintain genetic homogeneity between populations over long periods of time (Cano et al. 2008; Allendorf et al. 2010). The neutral SNP panel used in the present study provided evidence for significant but weak genetic differentiation between countries, in accordance with the findings of microsatellite markers, which displayed similar levels of genetic differentiation between Tasmania and New Zealand ($F_{ST} = 0.029$) (Morgan et al. 2013).

Performance of the outlier SNP panel for detecting population structure and implications for fisheries management

In the present study, the much higher and statistically significant F_{ST} value exhibited by the outlier SNP panel ($F_{ST} = 0.134$) demonstrated that loci putatively under selection have high power for detecting genetic structure between Australia and New Zealand and therefore could be used for determining country of origin in *J. edwardsii*.

Molecular techniques for seafood authentication are increasingly used to monitor fish stocks that are still commercially viable but are becoming threatened by overfishing, and to protect the consumer from fraudulent practices (Ogden 2008; Sorenson et al. 2013; Larraín et al. 2014). DNA barcoding is one of the preferred techniques due to its applicability to degraded material, low DNA requirement, simple protocol, time efficiency and reproducibility (Wong and Hanner 2008). However, DNA barcoding is more effective for interspecific differentiation, since it targets the mitochondrial cytochrome *c* oxidase I (COI) gene which can be highly conserved between subpopulations of the same species (Ogden 2008). In contrast, techniques that target the nuclear genome provide the potential for intraspecies assignment to population of origin (Nielsen et al. 2012a).

Trials using microsatellite markers for food traceability have been carried out with limited success in other commercial marine species. For example, a panel of nine microsatellites produced up to 50 % of correct assignments to country of origin of the mussel *Mytilus chilensis* in the presence of a global F_{ST} lower than 0.042 (Larraín et al. 2014). These authors suggested that having more informative loci and using SNP markers could improve assignment success. In contrast, higher success (92 %) was obtained when combining 13 microsatellite markers with data from the mitochondrial control region to accurately assign individuals to ocean of origin in Atlantic and Pacific blue marlin stocks with low genetic differentiation ($F_{ST} < 0.01$) (Sorenson et al. 2013). However, the suitability of loci putatively under selection for more reliable assignment to population of origin has been recognized (Martinsohn and Ogden 2009; Nielsen et al. 2012a), and recent studies have successfully demonstrated the power of outlier SNPs for this purpose (Araneda et al. 2016).

The outlier SNP panel obtained in the present study demonstrated strong genetic differentiation of *J. edwardsii* between Australia and New Zealand ($F_{ST} = 0.134$). Based on these markers, complimentary high throughput technologies that rely on prior sequence information, such as target capture or loop-mediated isothermal amplification (LAMP assay) (Tomita et al. 2008) could be used to differentiate lobsters caught in New Zealand from those caught in Australian waters in order to avoid mislabeling of country of origin. In particular, LAMP assay amplifies specific regions of the DNA with high specificity, efficiency, rapidity and low cost for preparation and visualization of results (Tomita et al. 2008). This assay is being increasingly used for clinical diagnosis of infectious diseases in developing countries since it does not require expensive laboratory equipment (i.e., a thermal cycler) and it can be performed in 1 h (Parida et al. 2008). For the particular case of *J. edwardsii*, primers could be specifically designed to target regions of the genome containing the outlier SNPs with highest loadings identified in Fig. S2 that account for most of the divergence between Australia and New Zealand (Martinsohn and Ogden 2009).

Genetic structure detected using outlier loci could suggest differences at regions of the genome putatively subject to selection. This divergence could be due to local adaptation to environmental conditions or to post-settlement mortality of unsuited genotypes (Holt and Gaines 1992; Caley et al. 1996; Marshall et al. 2010). Local adaptation to environmental conditions will result in genetic divergence between populations in the presence of high self-recruitment (Holt and Gaines 1992; Sanford and Kelly 2011). Herein, we demonstrated very high levels of self-recruitment within each country using outlier loci. Also, post-settlement mortality of unsuited genotypes will preserve the local genetic pool by removing migrants that are not fit to survive under particular environmental conditions (DeWitt et al. 1998; Marshall et al. 2010). Currently, it is impossible to unravel the degree at which self-recruitment and post-settlement mortality are contributing to genetic differentiation of *J. edwardsii* between Australia and New Zealand and this is beyond the scope of this study.

Based on the high genetic divergence resulting from the outlier SNP panel, we hypothesize that local conditions may have helped shape patterns of genetic diversity within adaptive regions of the genome in these populations, as shown in other studies (see Corander et al. 2013; Fraser et al. 2014), and however, in the absence of any reference genes or transcriptome for *J. edwardsii* or closely related species, it is impossible to ascertain this. We can only speculate that differences in environmental conditions between sampling sites may be driving differences at the outlier loci. Empirical evidence suggests that *J. edwardsii* are adapted to specific local conditions, but there is also extensive evidence of very high phenotypic plasticity. For example, growth rates in this species are highly variable and are mainly determined by temperature, density and food availability (Annala and Bycroft 1985; Jeffs and James 2001). Site-specific differences in carapace coloration and growth rates of this species have been reported in Australia (Punt et al. 1997; McGarvey et al. 1999). Individuals inhabiting deep waters are white-colored due to a diet with low concentration of carotenoid pigments and lower nutritional value, which can also impact growth negatively (McGarvey et al. 1999). Translocation experiments of white-colored lobsters into shallow areas demonstrated a change in coloration, growth rates and body condition after 12 months of translocation (Chandrapavan et al. 2009; Chandrapavan et al. 2010; Green et al. 2010; Chandrapavan et al. 2011). Therefore, even when self-recruitment could help retain locally adapted genotypes, phenotypic plasticity can also act to promote growth of *J. edwardsii* in certain environments.

Greater density of SNPs, together with improved but as yet unavailable genetic resources for *J. edwardsii* or closely related species (such as reference genome and transcriptome datasets), would provide further insight into potential genetic evidence of adaption to local environments. Seascape genetics could also help coupling local environmental conditions to genetic distance in order to explain patterns of genetic divergence between populations (Giles et al. 2015; Saenz-Agudelo et al. 2015).

Conclusions

In this study, we used ddRADseq and Illumina MiSeq next-generation sequencing to explore the genetic connectivity of southern rock lobster populations from Australia and New Zealand. Data from two SNP panels are presented, from which a panel of 121 outlier markers have allowed us to identify clear genetic structure in *J. edwardsii* populations between the two countries as well as high levels of self-recruitment. In addition, highly significant F_{ST} values estimated from the outlier SNP panel could be indicative of local adaptation driving the genetic differentiation between countries. This is particularly important in a commercial species managed by different agencies. Therefore, the outlier SNP panel developed in the present study could be used to differentiate New Zealand from Australian lobsters and therefore be useful for food traceability. We believe that more extensive sampling, including sites along the whole distribution of the species, could identify source and sink regions within each country precisely, which would also help management decision-making in Australia and New Zealand. Finally, continued development of genomics resources, such as transcriptome sequencing, gene characterization and quantitative trait locus discovery, is needed in order to explore the link between genotype, phenotype and the environment.

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Data accessibility Reference loci sequences available through Dryad, doi:10.5061/dryad.5c960.

Conflict of interest All authors declare that they have no conflict of interest.

Ethical approval All applicable international, national and/or institutional guidelines for the care and use of animals were followed.

References

ABARE-BRS (2010) Australian Fisheries Statistics 2009. Canberra

Allendorf FW, Hohenlohe PA, Luikart G (2010) Genomics and the future of conservation genetics. Nat Rev Genet 11:697–709. doi:10.1038/nrg2844

Altschul SF, Madden TL, Schaffer AA, Zhang J, Zhang Z, Miller W, Lipman DJ (1997) Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res 25:3389–3402

Annala JH, Bycroft BL (1985) Growth rate of juvenile rock lobsters (*Jasus edwardsii*) at Stewart Island, New Zealand. N Z J Marw Fresh 19:445–455

Antao T, Lopes A, Lopes RJ, Beja-Pereira A, Luikart G (2008) LOSITAN: a workbench to detect molecular adaptation based on a *F*(st)-outlier method. BMC Bioinform 9:323. doi:10.1186/1471-2105-9-323

Araneda C, Larraín MA, Hecht B, Narum S (2016) Adaptive genetic variation distinguishes Chilean blue mussels (*Mytilus chilensis*) from different marine environments. Ecol Evol. doi:10.1002/ece3.2110

Beaumont MA, Nichols RA (1996) Evaluating loci for use in the genetic analysis of population structure. Proc R Soc B 263:1619–1626. doi:10.1098/Rspb.1996.0237

Benestan L, Gosselin T, Perrier C, Sainte-Marie B, Rochette R, Bernatchez L (2015) RAD genotyping reveals fine-scale genetic structuring and provides powerful population assignment in a widely distributed marine species, the American lobster (*Homarus americanus*). Mol Ecol 24:3299–3315. doi:10.1111/mec.13245

Booth JD, Phillips BF (1994) Early life history of spiny lobster. Crustaceana 66:271–294. doi:10.1163/156854094x00035

Bruce B, Griffin DA, Bradford R (2007) Larval transport and recruitment processes of southern rock lobster. CSIRO, Hobart

Caley MJ, Carr MH, Hixon MA, Hughes TP, Jones GP, Menge BA (1996) Recruitment and the local dynamics of open marine populations. Annu Rev Ecol Syst 27:477–500

Candy JR, Campbell NR, Grinnell MH, Beacham TD, Larson WA, Narum SR (2015) Population differentiation determined from putative neutral and divergent adaptive genetic markers in Eulachon (*Thaleichthys pacificus*, Osmeridae), an anadromous Pacific smelt. Mol Ecol Res 15(6):1421–1434. doi:10.1111/1755-0998.12400

Cano JM, Shikano T, Kuparinen A, Merliä J (2008) Genetic differentiation, effective population size and gene flow in marine fishes: implications for stock management. J Integr Field Sci 5:1–10

Catchen JM, Amores A, Hohenlohe P, Cresko W, Postlethwait JH, De Koning DJ (2011) Stacks: building and genotyping loci de novo from short-read sequences. G3 1:171–182. doi:10.1534/g3.111.000240

Chandrapavan A, Gardner C, Linnane A, Hobday D (2009) Colour variation in the southern rock lobster *Jasus edwardsii* and its economic impact on the commercial industry. N Z J Mar Freshw 43:537–545

Chandrapavan A, Gardner C, Green BS (2010) Growth rate of adult rock lobsters *Jasus edwardsii* increased through translocation. Fish Res 105:244–247

Chandrapavan A, Gardner C, Green BS (2011) Haemolymph condition of deep-water southern rock lobsters (*Jasus edwardsii*) translocated to inshore reefs. Mar Freshw Behav Physiol 44:21–32

Corander J, Majander KK, Cheng L, Merila J (2013) High degree of cryptic population differentiation in the Baltic Sea herring *Clupea harengus*. Mol Ecol 22:2931–2940. doi:10.1111/mec.12174

Danecek P, Auton A, Abecasis G, Albers CA, Banks E, DePristo MA, Handsaker RE, Lunter G, Marth GT, Sherry ST, McVean G, Durbin R, Group GPA (2011) The variant call format and VCFtools. Bioinformatics 27:2156–2158. doi:10.1093/bioinformatics/btr330

DeWitt TJ, Sih A, Wilson DS (1998) Costs and limits of phenotypic plasticity. Trends Ecol Evol 13:77–81. doi:10.1016/S0169-5347(97)01274-3

Do C, Waples RS, Peel D, Macbeth GM, Tillett BJ, Ovenden JR (2014) NeEstimator v2: re-implementation of software for the estimation of contemporary effective population size (Ne) from genetic data. Mol Ecol Res 14:209–214. doi:10.1111/1755-0998.12157

Eckman JE (1996) Closing the larval loop: linking larval ecology to the population dynamics of marine benthic invertebrates. J Exp Mar Biol Ecol 200:207–237. doi:10.1016/s0022-0981(96)02644-5

Ellegren H (2014) Genome sequencing and population genomics in non-model organisms. Trends Ecol Evol 29(1):51–63. doi:10.1016/j.tree.2013.09.008

Fraser DJ, Debes PV, Bernatchez L, Hutchings JA (2014) Population size, habitat fragmentation, and the number of adaptive variation in a stream fish. Proc R Soc B 281:20140370. doi:10.1098/rspb.2014.0370

Giles EC, Saenz-Agudelo P, Hussey NE, Ravasi T, Berumen ML (2015) Exploring seascape genetics and kinship in the reef sponge *Stylissa carteri* in the Red Sea. Ecol Evol 5:2487–2502. doi:10.1002/ece3.1511

Green BS, Gardner C, Linnane A, Hawthorne PJ (2010) The good, the bad and the recovery in an assisted migration. PLoS ONE 5:e14160. doi:10.1371/journal.pone.0014160

Hedgecock D, Pudovkin AI (2011) Sweepstakes reproductive success in highly fecund marine fish and shellfish: a review and commentary. Bull Mar Sci 87(4):971–1002. doi:10.5343/bms.2010.1051

Henning F, Lee HJ, Franchini P, Meyer A (2014) Genetic mapping of horizontal stripes in Lake Victoria cichlid fishes: benefits and pitfalls of using RAD markers for dense linkage mapping. Mol Ecol 23:5224–5240. doi:10.1111/mec.12860

Holt RD, Gaines MS (1992) Analysis of adaptation in heterogeneous landscapes—implications for the evolution of fundamental niches. Evol Ecol 6:433–447. doi:10.1007/Bf02270702

Jacobsen MW, Pujolar JM, Bernatchez L, Munch K, Jian J, Niu Y, Hansen MM (2014) Genomic footprints of speciation in Atlantic eels (*Anguilla anguilla* and *A. rostrata*). Mol Ecol 23:4785–4798. doi:10.1111/mec.12896

Jeffs AG, James PJ (2001) Sea-cage culture of the spiny lobster *Jasus edwardsii* in New Zealand. Mar Freshw Res 52(8):1419–1424. doi:10.1071/MF01064

Jeffs AJ, Gardner C, Cockcroft A (2013) *Jasus* and *Sagmariasus* species. In: Phillips B (ed) Lobsters: biology, management, aquaculture and fisheries, 2nd edn. Blackwell, Oxford, pp 259–288

Jombart T, Ahmed I (2011) Adegenet 1.3–1: new tools for the analysis of genome-wide SNP data. Bioinformatics 27:3070–3071. doi:10.1093/bioinformatics/btr521

Jombart T, Devillard S, Balloux F (2010) Discriminant analysis of principal components: a new method for the analysis of genetically structured populations. BMC Genet 11:94. doi:10.1186/1471-2156-11-94

Karlin S, Altschul SF (1990) Methods for assessing the statistical significance of molecular sequence features by using general scoring schemes. P Natl Acad Sci USA 87:2264–2268

Langmead B, Salzberg SL (2012) Fast gapped-read alignment with Bowtie 2. Nat Methods 9:357-U354. doi:10.1038/NMETH.1923

Larraín MA, Díaz NF, Lamas C, Uribe C, Araneda C (2014) Traceability of mussel (*Mytilus chilensis*) in southern Chile using microsatellite molecular markers and assignment algorithms. Exploratory survey. Food Res Int 62:104–110. doi:10.1016/j.foodres.2014.02.016

Lennon NJ, Lintner RE, Anderson S, Alvarez P, Barry A, Brockman W, Daza R, Erlich RL, Giannoukos G, Green L, Hollinger A, Hoover CA, Jaffe DB, Juhn F, McCarthy D, Perrin D, Ponchner K, Powers TL, Rizzolo K, Robbins D, Ryan E, Russ C, Sparrow T, Stalker J, Steelman S, Weiand M, Zimmer A, Henn MR, Nusbaum C, Nicol R (2010) A scalable, fully automated process for construction of sequence-ready barcoded libraries for 454. Genome Biol 11:R15. doi:10.1186/gb-2010-11-2-r15

Luikart G, Cornuet JM (1998) Empirical evaluation of a test for identifying recently bottlenecked populations from allele frequency data. Conserv Biol 12:228–237. doi:10.1111/j.1523-1739.1998.96388.x/epdf

Luikart G, England PR, Tallmon D, Jordan S, Taberlet P (2003) The power and promise of population genomics: from genotyping to genome typing. Nat Rev Genet 4:981–994. doi:10.1038/nrg1226

Marshall DJ, Monro K, Bode M, Keough MJ, Swearer S (2010) Phenotype-environment mismatches reduce connectivity in the sea. Ecol Lett 13:128–140. doi:10.1111/j.1461-0248.2009.01408.x

Martinsohn JT, Ogden R (2009) FishPopTrace—developing SNP-based population genetic assignment methods to investigate illegal fishing. Forensic Sci Int: Genet Suppl Ser 2:294–296. doi:10.1016/j.fsigss.2009.08.108

Mastretta-Yanes A, Arrigo N, Alvarez N, Jorgensen TH, Piñero D, Emerson BC (2015) Restriction site-associated DNA sequencing, genotyping error estimation and de novo assembly optimization for population genetic inference. Mol Ecol Resour 15:28–41. doi:10.1111/1755-0998.12291

McGarvey R, Ferguson GJ, Prescott JH (1999) Spatial variation in mean growth rates at size of southern rock lobster, *Jasus edwardsii*, in South Australian waters. Mar Freshw Res 50:333–342. doi:10.1071/Mf97172

McKenna A, Hanna M, Banks E, Sivachenko A, Cibulskis K, Kernytsky A, Garimella K, Altshuler D, Gabriel S, Daly M, DePristo MA (2010) The Genome Analysis Toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Res 20:1297–1303. doi:10.1101/gr.107524.110

Merchant S, Wood DE, Salzberg SL (2014) Unexpected cross-species contamination in genome sequencing projects. Peer J 2:e675. doi:10.7717/peerj.675

Milano I, Babbucci M, Cariani A, Atanassova M, Bekkevold D, Carvalho GR, Espiñeira M, Fiorentino F, Garofalo G, Geffen AJ, Hansen JH, Helyar SJ, Nielsen EE, Ogden R, Patarnello T, Stagioni M, Tinti F, Bargelloni L (2014) Outlier SNP markers reveal fine-scale genetic structuring across European hake populations (*Merluccius merluccius*). Mol Ecol 23:118–135. doi:10.1111/mec.12568

Morgan EMJ, Green BS, Murphy NP, Strugnell JM (2013) Investigation of genetic structure between deep and shallow populations of the Southern rock lobster, *Jasus edwardsii* in Tasmania, Australia. PLoS ONE 8:e77978. doi:10.1371/journal.pone.0077978

Morin PA, Luikart G, Wayne RK, The SNP Workshop Group (2004) SNPs in ecology, evolution and conservation. Trends Ecol Evol 19:208–216. doi:10.1016/j.tree.2004.01.009

Nielsen EE, Hemmer-Hansen J, Larsen PF, Bekkevold D (2009) Population genomics of marine fishes: identifying adaptive variation in space and time. Mol Ecol 18:3128–3150. doi:10.1111/j.1365-294X.2009.04272.x

Nielsen EE, Cariani A, Mac Aoidh E, Maes GE, Milano I, Ogden R, Taylor M, Hemmer-Hansen J, Babbucci M, Bargelloni L, Bekkevold D, Diopere E, Grenfell L, Helyar S, Limborg MT, Martinsohn JT, McEwing R, Panitz F, Patarnello T, Tinti F, Van Houdt JKJ, Volckaert FAM, Waples RS, Carvalho GR, Albin JEJ, Baptista JMV, Barmintsev V, Bautista JM, Bendixen C, Berge JP, Blohm D, Cardazzo B, Diez A, Espineira M, Geffen AJ, Gonzalez E, Gonzalez-Lavin N, Guarniero I, Jerome M, Kochzius M, Krey G, Mouchel O, Negrisolo E, Piccinetti C, Puyet A, Rastorguev S, Smith JP, Trentini M, Verrez-Bagnis V, Volkov A, Zanzi A, Consortium F (2012a) Gene-associated markers provide tools for tackling illegal fishing and false eco-certification. Nat Commun 3:851. doi:10.1038/Ncomms1845

Nielsen R, Korneliussen T, Albrechtsen A, Li Y, Wang J (2012b) SNP calling, genotype calling, and sample allese frequency estimation from new-generation sequencing data. PLoS ONE 7(7):e37558. doi:10.1371/journal.pone.0037558

Ogden R (2008) Fisheries forensics: the use of DNA tools for improving compliance, traceability and enforcement in the fishing industry. Fish Fish 9:462–472. doi:10.1111/j.1467-2979.2008.00305.x

Paetkau D, Slade R, Burden M, Estoup A (2004) Direct, real-time estimation of migration rate using assignment methods: a simulation-based exploration of accuracy and power. Mol Ecol 13:55–65

Palumbi SR (1994) Genetic divergence, reproductive isolation, and marine speciation. Annu Rev Ecol Syst 25:547–572. doi:10.1146/Annurev.Ecolsys.25.1.547

Paradis E (2010) pegas: an R package for population genetics with an integrated-modular approach. Bioinformatics 26:419–420. doi:10.1093/bioinformatics/btp696

Parida M, Sannarangaiah S, Dash PK, Rao PVL, Morita K (2008) Loop mediated isothermal amplification (LAMP): a new generation of innovative gene amplification technique; perspectives in clinical diagnosis of infectious diseases. Rev Med Virol 18:407–421. doi:10.1002/rmv.593

Peterson BK, Weber JN, Kay EH, Fisher HS, Hoekstra HE (2012) Double digest RADseq: an inexpensive method for *De Novo* SNP discovery and genotyping in model and non-model species. PLoS ONE 7:e37135. doi:10.1371/journal.pone.0037135

Piry S, Alapetite A, Cornuet JM, Paetkau D, Baudouin L, Estoup A (2004) GeneClass2: a software for genetic assignment and first-generation migrant detection. J Hered 95:536–539

Pritchard JK, Stephens M, Donnelly PJ (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959

Punt AE, Kennedy RB, Frusher SD (1997) Estimating the size-transition matrix for Tasmanian rock lobster, *Jasus edwardsii*. Mar Freshw Res 48:981–992. doi:10.1071/Mf97017

Rannala B, Mountain JL (1997) Detecting immigration by using multilocus genotypes. Proc Natl Acad Sci USA 94:9197–9920

Saenz-Agudelo P, Dibattista JP, Piatek MJ, Gaither MR, Harrison HB, Nanninga GB, Berumen ML (2015) Seascape genetics along environmental gradients in the Arabian Peninsula: insights from ddRAD sequencing of anemonefishes. Mol Ecol 24:6241–6255. doi:10.1111/mec.13471

Sanford E, Kelly MW (2011) Local adaptation in marine invertebrates. Ann Rev Mar Sci 3:509–535. doi:10.1146/annurev-marine-120709-142756

Shanks AL (2009) Pelagic larval duration and dispersal distance revisited. Biol Bull-Us 216:373–385

Shanks AL, Grantham BA, Carr MH (2003) Propagule dispersal distance and the size and spacing of marine reserves. Ecol Appl 13:S159–S169

Sorenson L, McDowell JR, Knott T, Graves JE (2013) Assignment test method using hypervariable markers for blue marlin (*Makaira nigricans*) stock identification. Conserv Genet Resour 5:293–297. doi:10.1007/s12686-012-9747-x

Teske PR, Sandoval-Castillo J, van Sebille E, Waters J, Beheregaray LB (2015) On-shelf larval retention limits population connectivity in a coastal broadcast spawner. Mar Ecol Prog Ser 532:1–12. doi:10.3354/meps11362

Thomas L, Bell JJ (2013) Testing the consistency of connectivity patterns for a widely dispersing marine species. Heredity 111:345–354. doi:10.1038/hdy.2013.58

Tomita N, Mori Y, Kanda H, Notomi T (2008) Loop-mediated isothermal amplification (LAMP) of gene sequences and simple visual detection of products. Nat Protoc 3:877–882. doi:10.1038/nprot.2008.57

Waples RS, Do C (2010) Linkage disequilibrium estimates of contemporary Ne using highly variable genetic markers: a largely untapped resource for applied conservation and evolution. Evol Appl 3:244–262. doi:10.1111/j.1752-4571.2009.00104.x

Wei KJ, Wood AR, Gardner JPA (2013) Population genetic variation in the New Zealand greenshell mussel: locus-dependent conflicting signals of weak structure and high gene flow balanced against pronounced structure and high self-recruitment. Mar Biol 160:931–949. doi:10.1007/s00227-012-2145-9

Wong EHK, Hanner RH (2008) DNA barcoding detects market substitution in North American seafood. Food Res Int 41:828–837. doi:10.1016/j.foodres.2008.07.005

Wood DE, Salzberg SL (2014) Kraken: ultrafast metagenomic sequence classification using exact alignments. Genome Biol 15:R46. doi:10.1186/gb-2014-15-3-r46

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